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Paleomagnetism of the Cretaceous Morelos and Mezcala Formations, southern Mexico

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Abstract

A paleomagnetic study of platform-facies carbonate rocks of the mid-Cretaceous Morelos Formation and deep-water carbonate rocks of the overlying Upper Cretaceous Mezcala Formation, sampled at Zopilote canyon, in Guerrero State, southern Mexico, indicates that their characteristic magnetization was acquired contemporaneously with folding of these rocks during the Late Cretaceous Laramide orogeny. The remanence carrier is interpreted to be magnetite, although other mineral phases of high coercivity carry recent secondary overprints. The overall mean is of Dec = 323.1° and Inc = 36.5° ($k = 162.7$; $\alpha_{95} = 2.7°$; $N = 18$ sites; 64% unfolding). Comparison with the North America reference direction indicates that this area has experienced a small, yet statistically significant, counterclockwise direction of 19.2 ± 4.0°. Similar rotations are documented in other localities from southern Mexico; rotations are linked to mid-Tertiary deformation associated with the left-lateral strike-slip fault system that accommodated motion of the Chortis and Xolapa blocks.
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1. Introduction

Previous paleomagnetic studies of the Albian–Cenomanian Morelos Formation in southern Mexico (e.g., Urrutia-Fucugauchi, 1988; Böhnel, 1999) have concluded that the characteristic magnetization of this unit is a primary remanence, one acquired during or soon after deposition. Previous studies, which focused on the tectonic evolution rather than on the magnetostratigraphy, have also concluded that the paleomagnetic directions in the Morelos Formation indicate stability of the region with respect to cratonic North America since mid-Cretaceous time. Urrutia-Fucugauchi (1988) also pointed out the possible existence of a reversed polarity interval within the Cretaceous Normal Superchron (CNS) based on several intermediate directions from a discrete bed, low in the Morelos sequence.

The CNS is a normal polarity interval of unusual long duration, from 118 to 84 Ma (Opdyke and Channell, 1996). It was proposed in the late 1960s based on the absence of marine anomalies in Cretaceous sea-floor (Helsley and Steiner, 1969), and was generally confirmed in studies of deep-sea sediments

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and Tethyan pelagic sequences (Alvarez et al., 1977). Nevertheless, a few reversed polarity data have been reported for this interval, without posterior reconfirmation from other places. Events of very short duration interval within the CNS would be highly valuable magnetostratigraphic markers.

No field tests are available for the timing of remanence acquisition in the Morelos Formation, and previous claims of tectonic stability are incompatible with observations of widespread Tertiary deformation in this region of southern Mexico, including large vertical-axis rotations in Tertiary volcanic, intrusive, and sedimentary rocks (Urrutia-Fucugauchi et al., 1987). In order to evaluate the presence of a reverse polarity interval within the Morelos Formation, and provide better definition of the paleomagnetic record of this unit, we have conducted a systematic paleomagnetic sampling of the Morelos Formation and the lower part of the overlying Mezcala Formation, in northern Guerrero State, southern Mexico (Fig. 1). Here we discuss the results of this study and their tectonic implications.

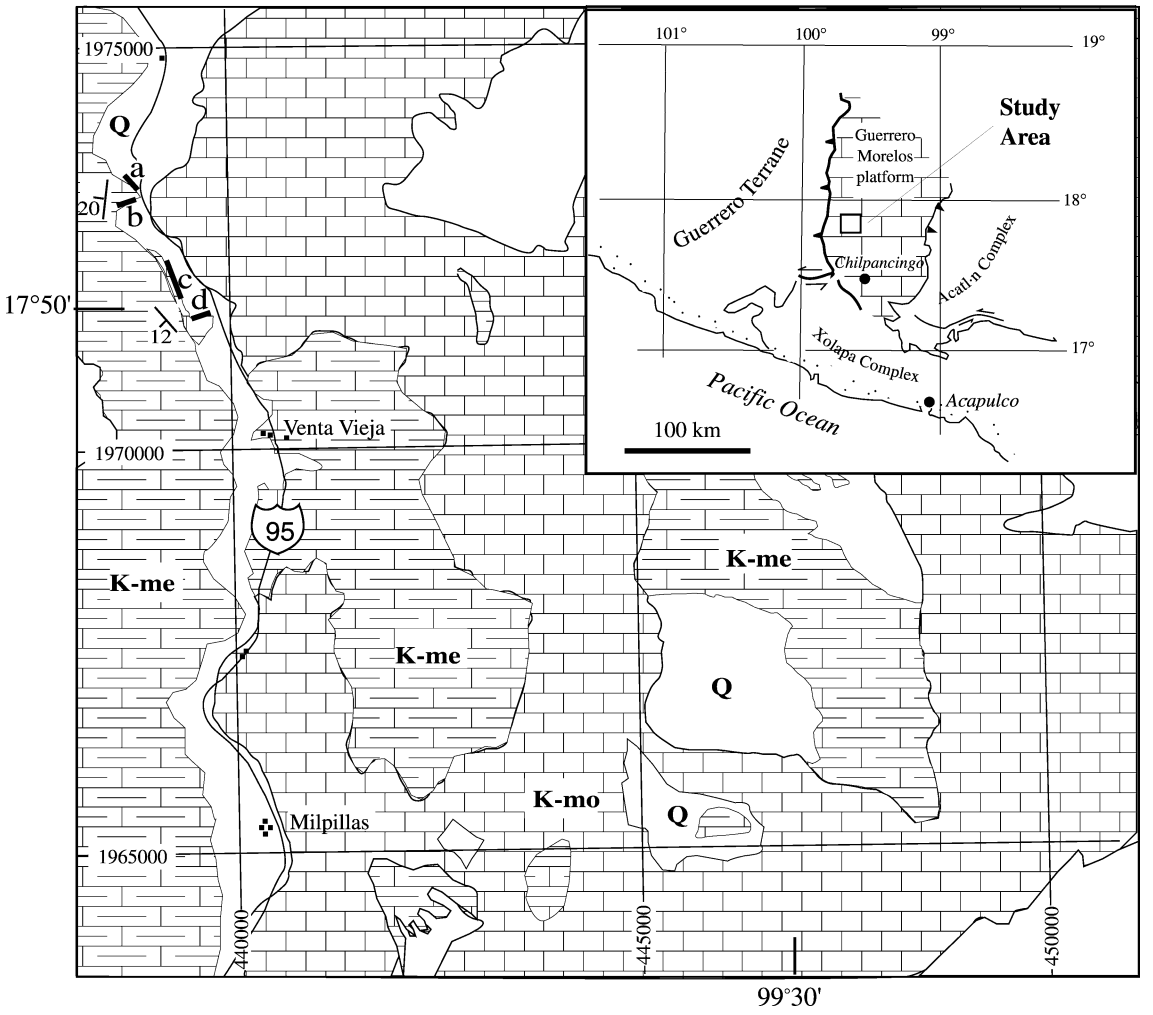


Fig. 1. Schematic geologic map of the Zopilote canyon area, northern Guerrero State, Mexico, showing paleomagnetic sampling localities along Highway 95. K-mo = Morelos Formation; K-me = Mezcala Formation; Q = Quaternary. Sections sampled (a–d) are illustrated in Fig. 2. The inset shows the tectonic setting of the Morelos–Guerrero platform in southern Mexico, thrusted from the west by rocks of the Guerrero terrane and truncated to the south by rocks of the Xolapa Complex.

2. Regional geologic setting and stratigraphy

The study area is characterized by a Cretaceous carbonate platform, the Guerrero–Morelos platform, developed on Paleozoic and Mesozoic basement rocks of the Mexican subcontinent (Fig. 1). Cretaceous strata are assigned, in ascending stratigraphic order to the Zicapa Formation, the Huitzuco Anhydrite, the Morelos Formation, and the Mezcala Formation (Fries, 1960). The Aptian (?) Huitzuco Anhydrite consists of sabkha deposits overlain or laterally replaced by a cyclothemic sequence assigned to the Zicapa Formation. This unit consists of conglomeratic sandstones, conglomerates, red beds and carbonate rocks of restricted distribution. The Albian–Cenomanian Morelos Formation (Fries, 1960) is a thick sequence of limestone and dolomite, mostly of lagoonal and platform facies. The Turonian–Maastrichtian Mezcala Formation consists of rhythmic intercalations of pelagic calcareous clays and limestone, which in their upper part are replaced by flysch-type sandstone and shale intercalations; Mezcala flysch is overlain in turn by a coarsening upward deltaic sequence of Maastrichtian age that includes some volcanic rocks assigned to the Tetelcingo Formation (Ortega-Gutiérrez, 1980). The Morelos–Mezcala contact in the study region is transitional, but abrupt. The contact is placed where shallow water carbonate sedimentation is replaced by carbonaceous shale and dark pelagic limestone, indicating rapid drowning of the platform. The causes of this drastic change in sedimentary conditions are relatively poorly understood.

Platform facies of the Morelos Formation consist of bioclastic packstone, wackestone and mudstone, with a relatively abundant microfauna with rests of Toucasias and pelecypods. Because of their shallow water character, rocks of the Morelos Formation are dominated by benthic formaminifer forms that limit biostratigraphic resolution. Biostratigraphic data for the Morelos Formation in northern Guerrero State indicate a Late Cenomanian age (Aguilera-Franco et al., 2001). The presence of *Nummoloculina heimi* and *Dyciclina schlumbergeri* about 200 m below the contact with the Mezcala Formation allows placing the Morelos Formation in the Upper Albian–Cenomanian. The range of *N. heimi* overlies the range of *T. primula* (Van Hinte, 1976; Castro Mora et al., 1975).

We know relatively little about sedimentological controls in the Guerrero–Morelos platform. The most recent works (Hernández-Romano et al., 1997; Aguilera-Franco et al., 2001) interpret facies in the western part of the platform, north of Chilpancingo and close to our study area, to indicate drowning of the platform near the Cenomanian–Turonian (C–T) boundary. These authors attribute this event to the C–T anoxic event, accompanied by tectonic subsidence.

Cretaceous rocks are folded along north to northeast trending structures. These structures are part of the Laramide-age fold and thrust belt that extends across north-central and eastern Mexico into the study area where it terminates abruptly against Tertiary strike-slip faults and the uplifted basement of the coastal Xolapa complex (Fig. 1). This Late Cretaceous compressional event has been linked to collision of one or several western-derived volcanic arcs, included in the Guerrero terrane of Campa and Coney (1983), with the platform and its hinterland (Tardy et al., 1994). The tectonic relationships between the volcanic terranes and the rest of southern Mexico have been interpreted as either allochthonous (Campa and Coney, 1983; Tardy et al., 1994), or as a paraauthochthonous arc developed on the continent margin (Elías-Herrera et al., 2000). Meschede and Frisch (1998) associate the collision of the Guerrero terrane with southern Mexico with similar events in Guatemala and the Greater Antilles. The absence of volcanic material in the Guerrero–Morelos platform suggests a distant paleogeography between the arc and the platform, but the presence of carbonate intercalations in the volcanic arc has been used as evidence of relative proximity between both paleogeographic elements (Cabral-Cano et al., 2000). Lang and Frerichs (1998) interpret rocks of the Mezcala Formation to record the initial pulse of the Laramide orogeny; based on Planktic foraminifera in Mezcala shales, they indicate a Coniacian age for this initial pulse. Naturally, deformation may have progressed from west to east as the colliding arc approached the platform.

Only recently, a large Tertiary event has been recognized in southern Mexico, which affects rocks of the Laramide fold-belt as well as Paleogene rocks (Meschede and Frisch, 1998; Cerca Martinez and Ferrari, 2001). This event is linked to truncation of the continental margin and the passage of the Chortis

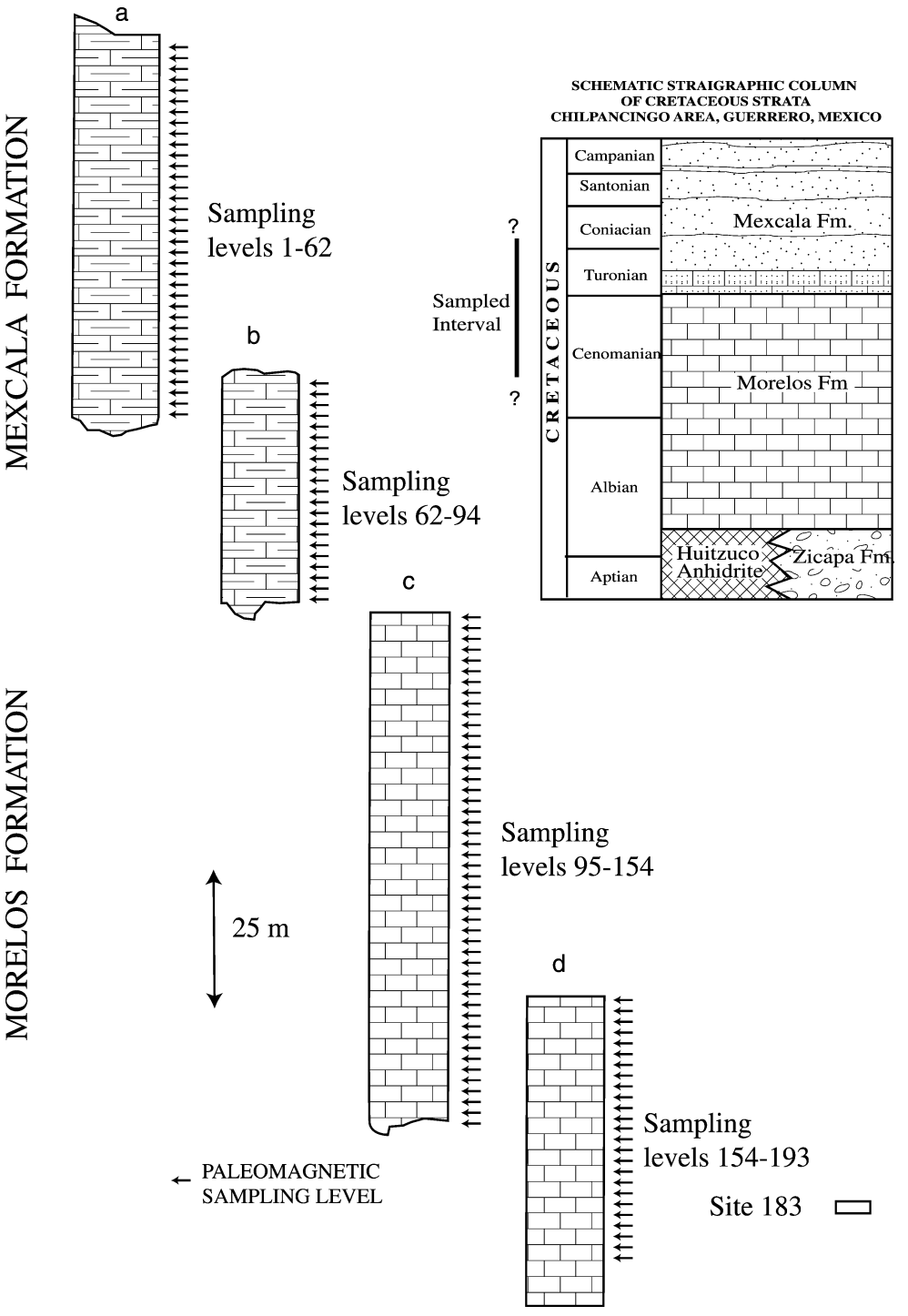


Fig. 2. Schematic stratigraphic section of the Morelos and Mezcala formation at Zopilote canyon, showing the distribution of sampling levels. The inset shows a simplified stratigraphic column of the region, after De Cserna et al. (1980).

block along the present location of the Mid-America trench. Tertiary deformation is evident in a wide belt parallel to the present coastline and extending from south of the Trans Mexican Volcanic Belt to the present North American–Caribbean plate boundary, the Polochic–Motagua fault zone. Eastward movement of the Chortis block along sinistral strike-slip faults is well documented in southern Mexico, where mylonites cut, or are intruded by, progressively younger plutonic rocks from west to east (Herrmann et al., 1994; Morán-Zenteno et al., 1996). South of the study area, the Late Eocene Tierra Colorada–El Ocotito granitoids (ca. 35 Ma) cut mylonites of the Tierra Colorada shear zone. Further south, Late Miocene (ca. 13 Ma) plutons are cut by mylonites in northwest Chiapas (Molina Garza et al., 1994) and a 2.8 Ma granite is cut by the Polochic fault near Motozintla in the Guatemala–Mexico border region. Transform faulting and associated uplift of the southern Mexico margin had a pronounced influence in Tertiary sedimentation and magmatism in southern Mexico.

3. Sampling

Samples were collected at Zopilote Canyon along the Acapulco–Mexico City Federal Highway (inset), in the same general region studied by Urrutia-Fucugauchi (1988). The Morelos and Mezcala formations are exposed in a broad antiform and dip gently west–northwest to southwest. Previously, a fold test in the Morelos Formation was not carried out because only a small section of the west limb of the broad antiform was sampled. Because the initial goal of the study was magnetostratigraphic, samples were collected in stratigraphic sequence, at intervals of 0.5 to 1.5 m, with one to three samples per interval (Fig. 2). The sampling scheme is somewhat unconventional by modern standards, but a sufficient number of cores was collected to assure reliability of the overall result. Calcite-filled veins and fine-grained iron oxides and hematite pigment bearing stylolites are common in thick beds of the Morelos Formation, but much less in the dark laminated limestone of the basal Mezcala. We collected a total of 194 samples, 94 in the Mezcala and 100 in the Morelos Formation, with a denser additional sampling in a bank found to contain magnetizations of reversed polarity within the Morelos Formation. Samples were drilled in the field and oriented with a magnetic compass. The total thickness sampled is about 300 m, in four slightly overlapping outcrops along the canyon walls (Figs. 1 and 2). The original study that reported reverse directions in the Morelos Formation (Urrutia-Fucugauchi, 1988) was based on the study of only 35 oriented samples from a ~ 30 m interval. Another, yet unpublished, study of the Morelos Formation at Zopilote canyon was part of a Master's thesis (Treviño-Rodríguez, 1986). It consisted of a study of 60 oriented samples from four short stratigraphic intervals. This study also reported reversed polarity directions within the Morelos Formation. The rather poor statistics of the site means reported in that study suggest that demagnetization experiments were relatively unsuccessful in isolating a characteristic magnetization, additionally affected by the low sensitivity of the Molspin magnetometer used for remanence measurements.

4. Paleomagnetism

4.1. Demagnetization results

Measurements of the natural remanent magnetization were carried out using a JR-5 spinner magnetometer, except for a few samples of weaker intensities which were measured using a 2G superconducting cryogenic magnetometer. In pilot studies, about 20% of the samples were subjected to thermal demagnetization in 12 steps up to 595 °C, and alternating field demagnetization (AF) in 10 to 15 steps, up to 150 mT. A simplified treatment of 5 or 6 thermal steps (NRM, 300, 350, 400, 450, and 500 °C) was used in the remaining samples, as thermal demagnetization proved more effective removing spurious overprints. Remanence components were identified in orthogonal demagnetization plots (Zijderveld, 1967) and their directions determined using standard principal component analysis techniques (Kirschvink, 1980).

NRM intensities are relatively high in the Mezcala Formation, of the order of 1–10 mA/m, and of intermediate to low values in the Morelos Formation (ranging from 0.1 to 2 mA/m). Demagnetization behavior of Mezcala and Morelos samples is similar.

The characteristic magnetization (ChRM) has moderate and distributed unblocking temperature (<590 °C; Fig. 3a and b). Maximum unblocking temperatures rarely exceed 500 °C, but demagnetization behavior is generally erratic above this temperature. Although AF appears to isolate a well-defined ChRM, similar to that isolated by thermal demagnetization, it is often evident that a large fraction of the remanence remains after treating the samples with 150 mT inductions (Fig. 3c and d).

The ChRM is northwest directed and of moderate inclination. It is overprinted by a soft and low unblocking temperature magnetization with positive inclination, likely of recent origin, that is removed with heating to ~200 °C (Fig. 3b). A marked decay of the remanence after heating to 100 °C in a number of samples is interpreted as indication that, locally, goethite carries part of the secondary overprint. The general orientation of the overprint suggests that it may be associated with recent weathering (Fig. 3f) and/or acquisition of a viscous remanence. In some cases, spurious overprints with aberrant directions were removed before the north-directed magnetization. The presence of goethite suggests that AF may not completely isolate the ChRM; thermal demagnetization was thus applied to the bulk of the collection.

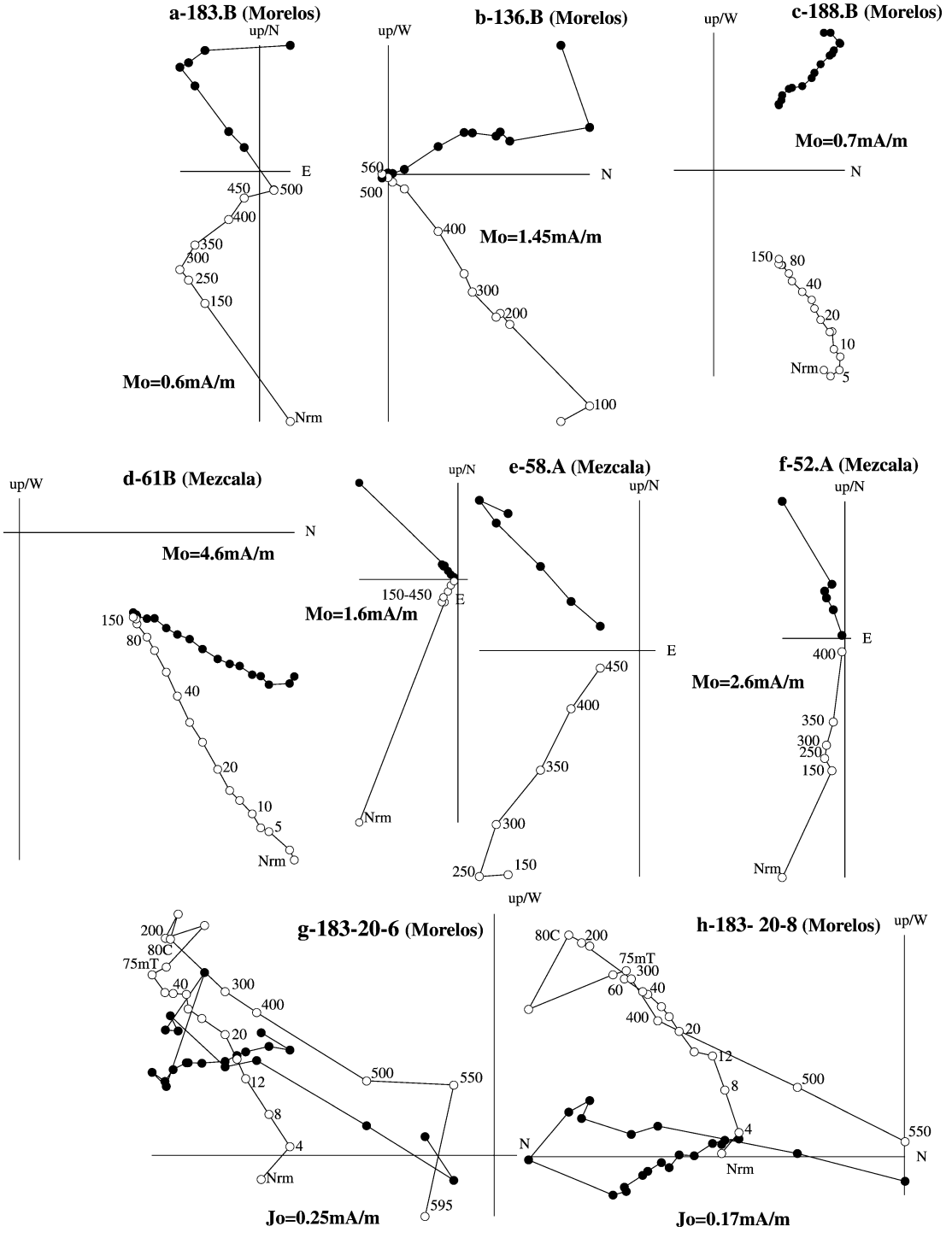
As mentioned above, a ~30 cm bed, located in the lower part of the Morelos Formation, carries a different magnetization (bed 183 in Fig. 2). The ChRM of this bed is also of distributed unblocking temperatures, between 200 and 590 °C, but its coercivity is much higher than in the rest of this unit (Fig. 3g and h). The ChRM of this bed is south directed and of moderate negative inclination. This magnetization is overprinted by a soft north-directed overprint (Fig. 3g), most of which is removed with inductions of ~60 mT. In either in-situ or tilt corrected coordinates, the reverse polarity magnetization is not antipodal to the magnetization observed in the rest of the collection (Fig. 4d). Its mean direction, based on only nine samples (Table 1), falls near the expected late Tertiary direction but data are insufficient to make an interpretation of its age possible.

4.2. Fold test

A fold test was applied to the ChRM of the section sampled; the test demonstrates that the remanence is contemporaneous with folding of the rocks. The data were analyzed using the method described in McFadden and Jones (1981) modified by McFadden (1990), and with the new fold test described in McFadden (1990). Individual specimens for the same sample were first combined. Then, samples for the same "limb" were averaged using conventional statistics (Fisher, 1953); the limb attitude is also defined as the average of a minimum of four attitude readings. A limb was defined as a contiguous set of beds with nearly identical attitude. Dip directions range from northwest (330°) to west–southwest (211°) and dips range from 6° to 36°, except for a site in the Mezcala with a dip of 53° (Table 1). The small variations in attitude within a limb are attributed to random errors caused by irregularities in the bedding surface or measurement mechanics (confidence intervals of the average attitude are typically less than 3°). The reverse polarity magnetization, which is not antipodal to the normal polarity direction, was excluded from the fold test. All data are summarized in Table 1. Samples were thus grouped into 18 "fold-limbs"; data were also analyzed for the Mezcala (10 limbs) and Morelos (8 limbs) subsets.

From an incremental fold test, it is first evident that the precision parameter k is a minimum in in-situ coordinates ($k=48.7$) and a maximum at ~60% partial unfolding (162.7). The incremental fold test is illustrated in Fig. 5, and is suggestive of a synfolding magnetization. Even by sampling a greater range of bedding attitudes, the fold test is not straightforward. The test of McFadden (1990) for limbs with unequal precision parameter gave the following results. When all 18 limbs were combined, the maximum of the V statistic occurs at 0% unfold-

Fig. 3. Orthogonal demagnetization diagrams for rocks of the Morelos (a–c) and Mezcala (d–e) formations (in geographic coordinates). Samples for diagrams (c) and (d) were subjected to AF demagnetization. Diagrams (g) and (h) correspond to samples for an interval of reverse polarity low in the Morelos Formation and discussed in the text. These samples were subjected to AF and thermal demagnetization. Solid (open) symbols are projections into the horizontal (vertical) plane.


a-183.B (Morelos)
up/N
E
450
500
400
350
300
250
150
Mo=0.6mA/m
Nrm
b-136.B (Morelos)
up/W
N
560
500
400
300
200
100
Mo=1.45mA/m
c-188.B (Morelos)
up/W
N
Mo=0.7mA/m
150
80
40
20
10
Nrm
5
d-61B (Mezcala)
up/W
N
Mo=4.6mA/m
150
80
40
20
10
5
Nrm
e-58.A (Mezcala)
up/N
150-450
E
Mo=1.6mA/m
Nrm
up/N
E
450
400
350
300
250
150
f-52.A (Mezcala)
up/N
E
400
Mo=2.6mA/m
350
300
250
150
Nrm
g-183-20-6 (Morelos)
up/W
200
80C
75mT
40
300
400
20
12
500
550
8
4
Nrm
595
Jo=0.25mA/m
h-183- 20-8 (Morelos)
up/W
80C
200
75mT
300
60
40
400
20
12
8
500
4
550
N
N
Nrm
Jo=0.17mA/m

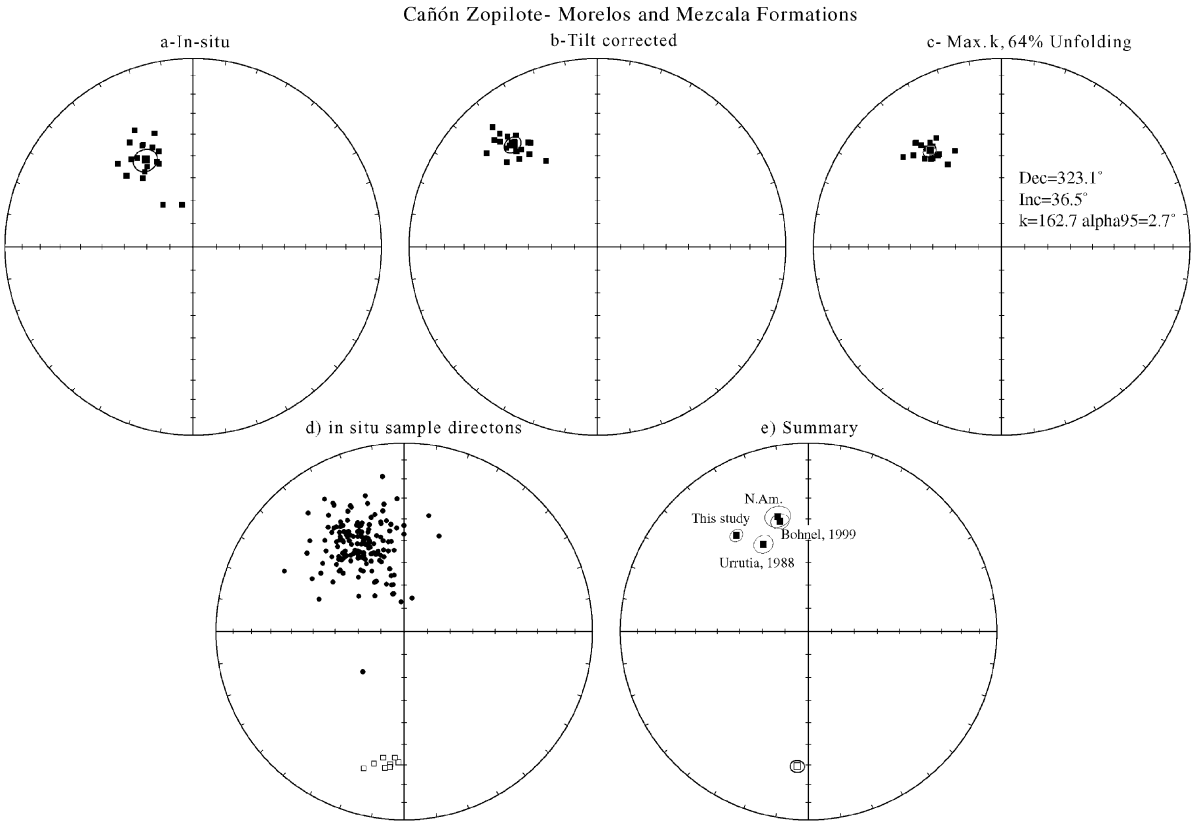


Fig. 4. Equal-area stereographic projections of paleomagnetic data for the Morelos and Mezcala formations. (a) In situ mean directions for the 18 "limbs" combined for the fold test; (b) tilt corrected directions; (c) mean directions that maximize the precision parameter *k* (at 64% unfolding); (d) individual sample directions (in in-situ coordinates); (e) summary of paleomagnetic directions for the Morelos Formation compared to the reference expected direction for the North American craton.

ing ($V=766.75$; Table 1), the minimum V statistic occur at 70% unfolding ($V=112.8$), and V is 191.08 in tilt corrected coordinates; the test, however, is not significant at the 95% confidence level (the critical value of V is 92.1). A similar result (an inconclusive fold test) is obtained for the Mezcala subset. In the case of the Morelos subset, the value of the V statistic indicates that (with a 95% confidence) the null hypothesis of a common mean to all limbs can be rejected in the in-situ and tilt-corrected positions, but not in an intermediate position (58% unfolding). This was already apparent in Fig. 5.

An alternative fold-test designed by McFadden (1990) seeks correlation (or lack thereof) between limb-paleomagnetic directions and limb-attitudes. The test uses the SCOS statistic for this purpose, and a high value of the statistic is interpreted to indicate strong correlation (a negative fold test). For the combined set of 18 limbs, there appears to be correlation in in-situ and tilt corrected coordinates, but with a 95% confidence, no correlation is observed at the maximum k value (64% unfolding). For the Mezcala subset, no apparent correlation is observed in tilt-corrected or at maximum k value (74% unfolding); that is, the apparent synfolding result of a fold test in the Mezcala subset is not statistically significant at the 95% confidence level. The Morelos "limbs" give again a statistically significant synfolding magnetization, at ~ 47% unfolding. The magnetization is thus secondary in the Morelos Formation, and apparently so in the Mezcala. Considering the Mezcala Formation has been interpreted as synorogenic, it is possible that it acquired its magnetization when beds were near horizontal but Morelos lime-

Table 1
Paleomagnetic data and statistical parameters

	n	Dec (°)	Inc (°)	*k*	α_{95}	Attitude	
Morelos Fm.							
Limb 1	42	337.9	50.5	67.2		282, 30	
Limb 2	6	337.3	49.1	96.4		280, 22	
Limb 3	7	333.5	39.6	40.5		266, 33	
Limb 4	5	340.5	45	40.6		254.5, 33	
Limb 5	7	337.8	42.3	45.7		250, 23	
Limb 6	4	341.3	36.3	47.9		211, 16.5	
Limb 7	4	328.8	35.4	446.2		241, 13	
Limb 8	8	333.5	31.3	54.4		227, 11	
	Limbs					*V*-stat.	SCOS2
In Situ	8	336.1	41.2	116.8	5.1	125.120	3.508
Tilt corrected	8	320.4	34.0	104.1	5.5	42.766	4.300
Max. *k* stat. (47% unfolding)	8	328.0	38.6	221.9	3.7		1.280
Max. *R* stat. (58% unfolding)	8	326.2	37.7			25.482	
				95% Critical value =		33.298	3.298
Mezcala Fm.							
Limb 1	7	313.0	47.1	188.2		330, 23	
Limb 2	4	324.9	42.4	99.2		303.5, 18	
Limb 3	4	317.9	40.1	95.7		278, 20	
Limb 4	5	330.4	49.2	583.1		307, 23	
Limb 5	4	328.0	43.5	106.9		283, 6.5	
Limb 6	4	327.4	50.6	143.2		303, 36	
Limb 7	4	324.1	52.7	175.4		303, 24.5	
Limb 8	9	345.4	71.2	233.9		307, 53	
Limb 9	4	324.7	67.6	183.4		330, 33	
Limb 10	29	334.0	39.3	63.8		267, 20	
						V-stat.	SCOS2
In situ	10	326.0	50.6	45.3	7.3	526.630	5.909
Tilt corrected	10	318.3	29.8	114.1	4.5	111.738	2.936
Max. *k* stat. (72% unfolding)	10	321.1	33.9	151.3	3.9		0.52
Max. *R* stat. (79% unfolding)	10	320.8	32.2			81.078	
				95% Critical value =		42.09	3.685
Combined						*V*-stat.	SCOS2
In situ	18	331.0	46.5	48.7	5	766.746	14.463
Tilt corrected	18	320.2	30.3	96.2	3.5	191.08	8.715
Max. *k* stat. (64% unfolding)	18	323.1	36.5	162.7	2.7		2.130
Max. *R* stat. (70% unfolding)	18	322.6	35.4			112.83	
				95% Critical value =		71.31	4.940
Morelos Fm.	28	332.7	46.8		4.0	(Urrutia-Fucugauchi, 1988)	
Reverse polarity interval	1\9	181.2	−29.9	333.8	2.8		

Here, *n* is the number of samples for each of the "limbs" to which samples were assigned. Dec and Inc are the declination and inclination of the mean direction, and *k* and α_{95} are Fisher's statistical parameters. For the fold test we use the *V* and SCOS2 statistics, after McFadden (1990). Attitudes are described by dip azimuth and dip angle.

stones had already been partially folded. We conclude, however, that the magnetization is secondary throughout the section.

One concern is that the apparent synfolding magnetization could be the result of superposition of two deformation events, a complex folding mechanism, or internal deformation (Kodama, 1986; Van der Plujim, 1987). Evidence of superposition of bedding rotations is derived from a high dispersion in declination (apparent tectonic rotations; MacDonald, 1980). The

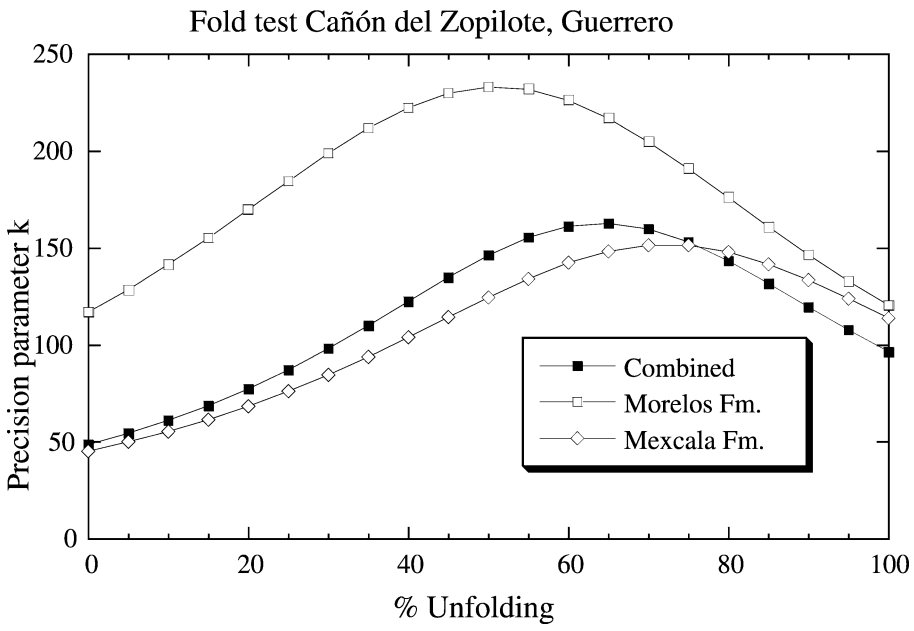


Fig. 5. Incremental fold test for samples collected at Zopilote canyon in the Morelos and Mezcala formations.

distribution of limb-means does not suggest that declinations are more scattered than inclinations (Fig. 4d), and we find no field evidence of significant internal deformation. Furthermore, bedding dip angles are small and evidence of a complex folding mechanism is lacking. We thus argue that the synfolding result indicates acquisition of a secondary remanence during the Laramide orogeny (with the local use of the term indicating a Late Cretaceous compressional event). The near uniform polarity of the ChRM suggests remagnetization during a long episode of normal polarity. The Laramide orogeny in southern Mexico ended in Maastrichtian time, since lamprophyres in the Mezcala mining district are undeformed

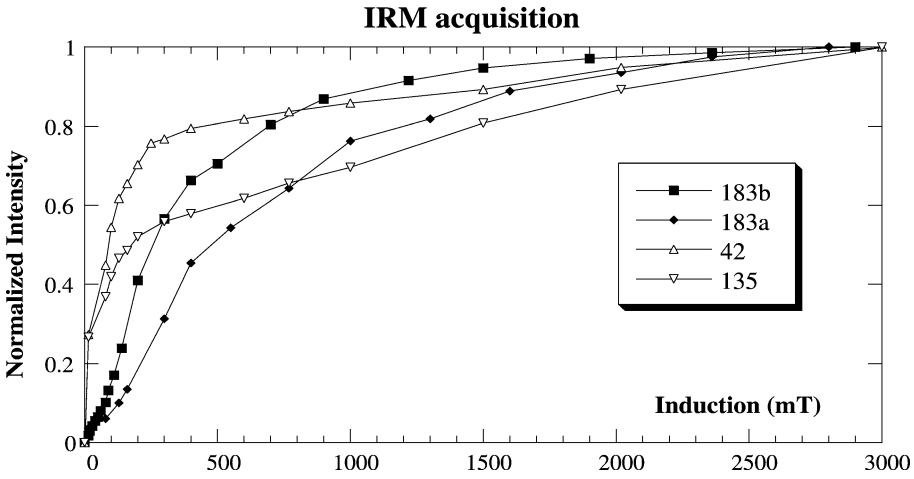


Fig. 6. Isothermal remanent acquisition experiments for selected samples; samples 183a and 183b are from the interval of reversed polarity in the Morelos Formation.

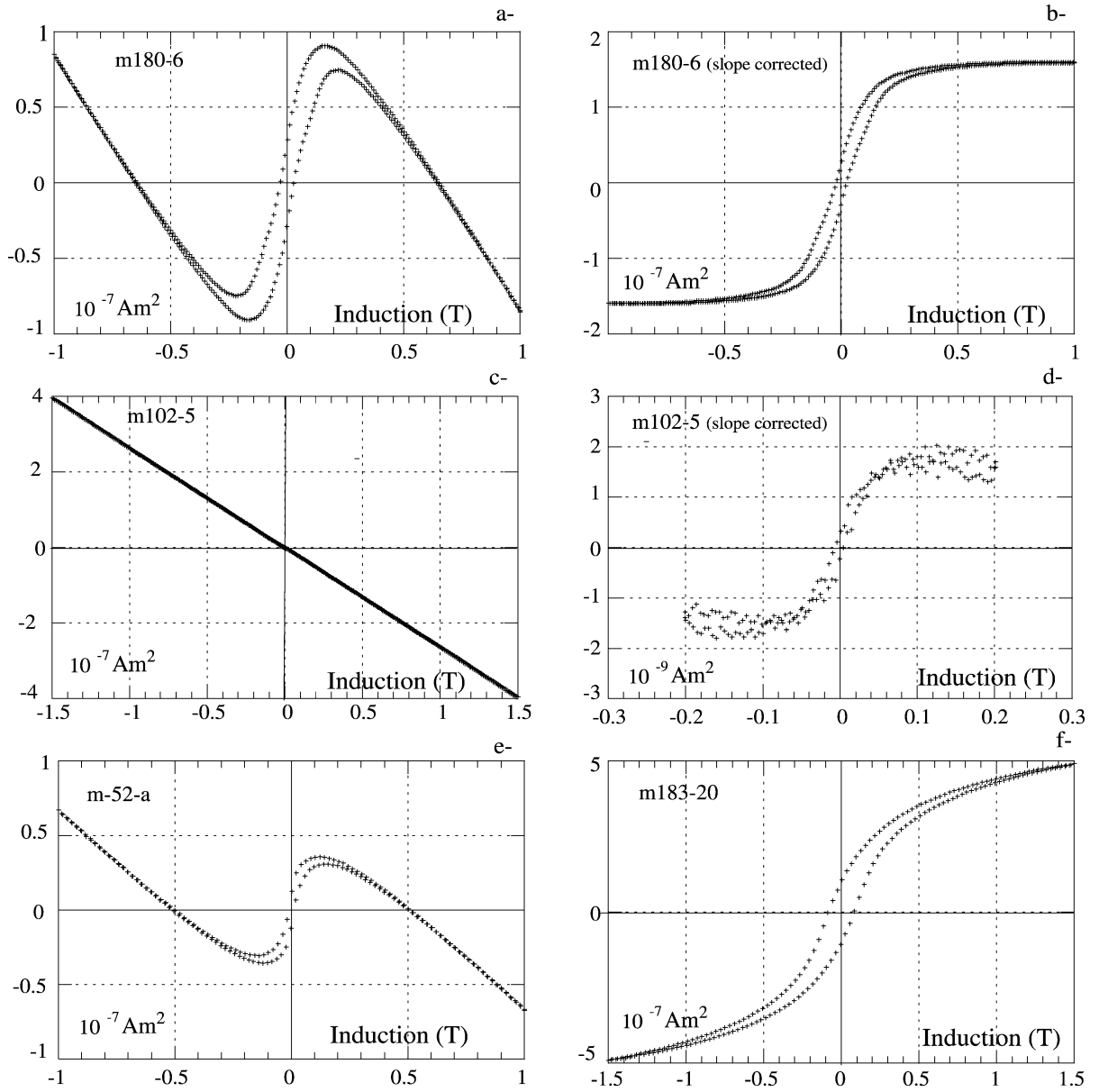


Fig. 7. Typical hysteresis loops of samples of the Morelos Formation. The loops show in general a strong contribution of diamagnetic minerals (a, c, and e). Additionally, there are varying contributions of ferromagnetic particles. We show in diagrams (b) and (d) the slope corrected diagrams of corresponding samples (a and c). In sample M102 ferromagnetism is very low, in good agreement with the low NRM intensity of this sample. Samples with higher NRM intensities (e.g., M180-6) are characterized by higher saturation magnetization and coercivities a few tens of mT, thus typical for small magnetite or maghemite particles. Samples from the reversed bed (M183), which suggest macroscopically and microscopically the presence of hematite, do not saturate in inductions of 1.5 T and are characterized by much higher coercivities of ~ 100 mT. Diagrams (e) and (f) are not corrected for the slope of diamagnetic or paramagnetic phases.

and give radiometric ages in the range of 63 to 65 Ma (Jones and Jackson, 1999). This is thus interpreted as the minimum age of remagnetization. The timing of remanence acquisition is thus bracketed between Coniacian and Maastrichtian time.

Evidence of remagnetization is widespread in the Laramide fold and thrust belt of northeastern Mexico (Böhnel et al., 1990; Nowicki et al., 1993). Our data suggest that remagnetization affected the southern portion of the fold belt as well.

5. Rock magnetism and petrography

Rock magnetic studies indicate the presence of mineral mixtures; magnetite (or maghemite) is the dominant mineral in the samples, although hematite and goethite are also present. The acquisition of IRM reveals the presence of a magnetic phase of low coercivity, evident by a steep initial increase at low inductions, and a high coercivity phase(s) that does not saturate with inductions of 3 T (Fig. 6). Urrutia-Fucugauchi (1988) had previously suggested saturation with inductions of 0.3 T, but that is evidently not the case (see their Fig. 4). There is a clear contrast between the IRM curves of the thin interval of reverse polarity and the rest of the collection. At high inductions, typical samples of the Morelos and Mezcala formations have gently concave-upwards curves (indicative of goethite); the initial steps increase in remanence is also faster than for samples of the reverse polarity bed. Furthermore, samples of the reverse polarity bed appear to approach saturation with inductions of 3 T (suggesting the presence of hematite, rather than goethite).

Based on hysteresis measurements we distinguish three different behaviors or groups of samples. One has a clear ferromagnetic signal, with loop parameters typical of PSD grain sizes (Fig. 7b). The second group consists of samples dominated by diamagnetism and an almost insignificant contribution from ferromagnetic material (Fig. 7c,d). A relatively small group of samples exhibit very high coercivities (Fig. 7f). This group is also characterized by hematite pigment (red coloration) along fractures and stylolites. None of the samples exhibit the characteristic wasp-waisted loops typical of iron oxides precipitated from a fluid phase and SD–SP mixtures.

6. Tectonic implications

The overall mean directions of the ChRM for samples of the Mezcala and Morelos formations are of Dec = 321.1° and Inc = 33.9° (maximum $k = 151.3$, at 72% unfolding) and Dec = 328.0° and Inc = 38.6° (maximum $k = 221.94$, at 47% unfolding), respectively (Table 1). The combined set of 18 "limbs" gives an overall mean of Dec = 323.1° and Inc = 36.5° (maximum $k = 162.7$, at 64% unfolding). This direction differs significantly form the mean reported by Urrutia-Fucugauchi (1988, Fig. 4e). Because the mean direction reported here is based on a larger collection and no fold test was presented in the previous study of the Morelos Formation, we believe that the revised mean of the ChRM is a more accurate estimate of the paleofield direction in Late Cretaceous time. This direction yields a paleomagnetic pole at 55.1°N–180.7°E, which falls off the reference apparent polar wander path of North America.

The expected direction for the sampling site assuming stability with respect to the craton, calculated for a reference cratonic pole at 72.3°N–194.7°E (after McElhinny and McFadden, 2000), has Dec = 342.3° and Inc = 42.0°; the reference pole includes data for the interval from 97 to 67 Ma, thus closely resembling the estimated age range during which remanence must have been acquired. A major finding is, therefore, that the observed declination is clearly discordant (Fig. 4e). When the observed mean declination and expected declination are compared, we obtain a counterclockwise rotation of the sampling site of 19.2 ± 4.0°; inclinations indicate a flattening value of 5.5 ± 4.4°. Although the flattening value is statistically significant, latitudinal movement of the site is indeed uncertain. We note that the mean for Paleocene poles listed in McElhinny and McFadden (2000) is located at 75.0°N–168.4°E; using this pole, the expected inclination for the sampling site is 31.0°. Comparing this value with the 42.0° of the Late Cretaceous reference, it is evident that sufficient uncertainty exists in the timing of remanence acquisition such that had we compared the Morelos platform result ($I = 36.5°$) to a pole intermediate between the Paleocene and Late Cretaceous poles, the flattening value would be insignificant. The conclusion of a statistically significant counterclockwise rotation is, nonetheless, inescapable.

It is important to demonstrate whether the counterclockwise rotation observed at Cañón Zopilote is of regional scale, and thus related to plate motions, or it is of limited scale and thus related to local deformation. Solving this question requires an analysis of the paleomagnetic database for southern Mexico.

The paleomagnetic database for southern Mexico is actually quite extensive, but average data quality is low, and data in unpublished theses is difficult to evaluate. A paleomagnetic pole is available for the Morelos Formation in western Oaxaca, about 200 km of the study area (Böhnel, 1999) where it overlies Jurassic rocks and the Paleozoic basement of the Acatlán Complex. A fold test is not available, although the characteristic magnetization was interpreted as a primary remanence (Böhnel, 1999); the mean direction (Fig. 4e) is concordant in tilt-corrected coordinates, but gently rotated counterclockwise in in-situ coordinates (Table 2).

Paleomagnetic data are also available for mid-Cretaceous though Paleogene volcanic and plutonic rocks of the coastal Guerrero terrane (Böhnel et al., 1988, 1989; Table 2). Because no reference to paleo-horizontal is available in plutonic rocks, the use of these data for tectonic interpretations is cautioned. Most relevant to our discussion are results for the Puerto Vallarta intrusive complex and the El Encino, Autlán and Zihuatanejo areas, for which limited data for sedimentary and/or volcanic rocks are available, and data are of acceptable quality (Fig. 8, Table 2). All data for the El Encino area were grouped in view of the fact that they fail a fold test and may be qualitatively linked to remagnetization by the nearby Pihuamo Paleocene plutonic complex (Pantoja-Alor, 1983).

It is evident from Table 2 that several rock units indicate moderate amounts of counterclockwise rotation, with rotation estimate values between about 7° and 25°. Only the Puerto Vallarta intrusive complex and the Morelos Formation in Oaxaca yield concordant declinations. In contrast, flattening values are small to insignificant. In Table 2, Tertiary data for the Balsas Group (Urrutia-Fucugauchi, 1983) and for mid-Tertiary plutons are of lesser quality than for the more coastal localities listed above. These suggest similar rotation values (~ 12° to 18°) and are included here for the sake of completeness. However, no reliable data are available for Eocene and younger time. Fig. 8 illustrates major tectonic elements and the distribution of paleomagnetically determined rotations.

Based on paleomagnetic data for the region, Böhnel et al. (1989) concluded that the continental margin of Mexico has experienced localized tectonic rotations. New data for the Morelos and Mezcala formations as well as a reassessment of previously published data from southern Mexico suggest that counterclockwise rotations are widespread and to some degree systematic. This in turn suggests that a common mechanism may be responsible for the observed rotations. As mentioned above, there is ample evidence along the continental margin of a left lateral strike-slip regime recording eastward motion of the Chortis and Xolapa blocks. Latest Cretaceous (?) to mid-Tertiary mylonites are grossly parallel to the continental margin, and they occur over a distance >300 km from the coast. We thus interpret paleomagnetic data for this region to indicate counterclockwise vertical-axis rotations of crustal blocks of regional scale due to interplate stress transmission and shear between major strike-slip faults. Inspection of Fig. 8 suggests that other mechanisms for block rotation may be drag near major faults (see the change in Laramide trends near the Tierra Colorado shear zone, Fig. 8) and pull-apart zones (possibly illustrated north of Taxco in the Chontalcuatlán basin, Alaniz-Álvarez et al., 2002).

Rotations in the Puerto Vallarta and Manzanillo areas of Fig. 8 are small to negligible. This is to be expected. Because no rapid mid-Tertiary uplift, nor mylonite belts, have been reported in these areas, it has been assumed that the Chortis Block was located south of these localities (Morán-Zenteno et al., 1996). Thus, strike-slip tectonics are limited to latitudes south and east of the Colima graben.

Jurassic rocks in Oaxaca have been used to constrain the paleogeography and tectonic evolution of southern Mexico (e.g., Ortega-Guerrero and Urrutia-Fucugauchi, 1993; Böhnel, 1999; Morán-Zenteno et al., 1988). In general, paleomagnetic data for Jurassic rocks yield discordant directions, but the timing of remanence acquisition is difficult to ascertain. In Table 2, we summarize relevant Jurassic data from southern Mexico. It is apparent that Mid to Upper Jurassic strata have characteristic magnetizations that resemble those observed in Cretaceous and younger rocks, either in in-situ or tilt-corrected coordinates (or, in cases, both). We suspect that these magnetizations may be secondary in origin. Declinations in Jurassic rocks, which are from localities east of the Morelos platform, are

Table 2
Jurassic–Tertiary paleomagnetic data fo the Mixteca and Guerrero terranes, southern Mexico

Rock unit	Rock age	Locality		Q							N	Dec-obs	Inc-obs	k	α_{95}	Mag-age	Dec-exp	Inc-exp	R	ΔR	F	Ref.	Comments
		°N	°W	1	2	3	4	5	6	7													
Balsas Group	Paleo.-Eo.	18.55	99.6	×	–	×	–	–	–	×	3	331.9	31.9	30.2	22.8	Eoc.	349.9	30.1	– 18.0	19.0	– 1.8	1[a]	tilt corrected
Xaltianguis Intrusive	30 ± 1	17.15	99.71	×	–	–	–	–	×	–	3	337.6	33.5	20.7	27.8	Oligoc.	352.9	24.8	– 15.3	23.6	– 8.7	9	
Acapulco Complex	43 ± 1	16.5	99.5	×	–	–	×	?	–	×	12	337.4	43.6	12.8	10.7	Eoc.	349.9	26.7	– 12.5	10.0	– 16.9	2	
Autlán volc. And intr.	Ku–Tl	19.5	104.5	–	×	×	×	×	–	–	14	337.1	34.7	21.8	8.7	Ku–Tl	344.3	35.4	– 6.9	8.1	0.7	3	tilt corrected
												(334.0)	(41.9)	15.9	10.3								in situ
El Encino Volc-Sed	Apt.–Mastr.	19.14	103.36	–	×	–	×	–	–	×	28	321.4	35.2	12.8	7.9	Paleoc.	344.3	34.4	– 22.9	7.5	– 0.8	3[a]	in – situ (sites 43–76)
Zihuatanejo Intrus.	44–61 Ma	18.12	102.20	×	–	–	×	?	–	×	11	322.9	32.5	7.1	18.5	Tl	347.0	30.9	– 24.1	15.9	– 1.6	3[a]	(sites 82–93)
Morelos Fm.	Alb.–Cen.	18.12	97.96	–	×	–	×	–	–	–	1 (27)	345.5	39.5	60.4	3.6	?	342.2	40.2	3.3	3.6	0.7	4	tilt corrected
												(330.4)	(38.9)	60.4	3.6		342.1	41.8	– 11.7	4.5	2.9		in-situ
Mezcala–Morelos Fm.	Cen–Tur.	17.83	99.56	×	×	–	×	–	×	×	18	323.1	36.5	162.7	2.7	San.–Mastr.	342.3	42.0	– 19.2	4.0	5.5	5	synfolding
Puerto Vallarta Complex	100–80 Ma	20.5	105.3	×	×	–	×	×	–	×	15	347.7	41.2	25.8	7.7	100–80?	342.6	45.9	5.1	7.0	4.7	3	
Suspected remagnetized Jurassic strat, Mixteca Terrane																							
Cidaris Limestone	Oxfordian	17.25	97.74	–	×	×	×	–	–	–	7	11.5	38.6	88.4	6.5							6[a,b]	tilt corrected
Yucuñuti Fm.	Callovian	17.66	97.83	–	–	×	×	–	–	–	3	329.8	34.8	68.8	15.0							6[b]	tilt corrected
Tecomazúchil Group	Mid. Jurassic	18.05	97.98	–	×	×	×	×	–	–	1 (27)	5.6	34.6	47.6	4.1							4	tilt corrected
												(354)	(42.3)	47.6	4.1							4	in-situ
Piedra Hueca Unit	Mid. Jurassic	18.27	97.8	–	–	×	×	–	–	–	3	346.9	24.5	79.5	13.9							7[a]	tilt corrected
												(344.4)	(8.5)	173.8	9.4								in-situ
Otlaltepec Unit	Up. Jurassic	18.3	97.8	–	×	×	×	×	–	–	6	355.9	3.2	24.8	12.4							7[a]	tilt corrected
												4.7	41.6	22.4	13.0								in-situ
Rosario Fm.	Mid. Jurassic	17.60	97.87	–	–	×	×	–	–	×	1 (15)	210.4	– 46.2	42.8	5.9							4	tilt corrected
												(231.6)	(– 51.9)	42.8	5.9							4	in-situ
Cratonic reference poles	Kl	72.7°N, 191.1°E													2.1							8	
	Ku	72.3°N, 194.8°E													3.7								
	Paleocene	75.2°N, 168.4°E													3.9								
	Eocene	80.0°N, 158.0°E													3.7								

References: 1—Urrutia-Fucugauchi (1983, 1980); 2—Böhnel et al. (1988); 3—Böhnel et al. (1989); 4—Böhnel (1999); 5—This study; 6—Morán-Zenteno et al. (1988); 7—Ortega-Guerrero and Urrutia-Fucugauchi (1993); 8—McElhinny and McFadden (2000); 9—Urrutia-Fucugauchi and Molina-Garza (1995). Q, reliability: (1) Age of magnetization, (2) Statistics of sufficient quality, (3) Reference to paleohorizontal, (4) Detailed demagnetization data, (5) Dual polarity magnetizations; (6) Field test, (7) Resembles younger data.

[a] Poles recalculated here.

[b] Original publication does not provide sufficient data to calculate in-situ directions.

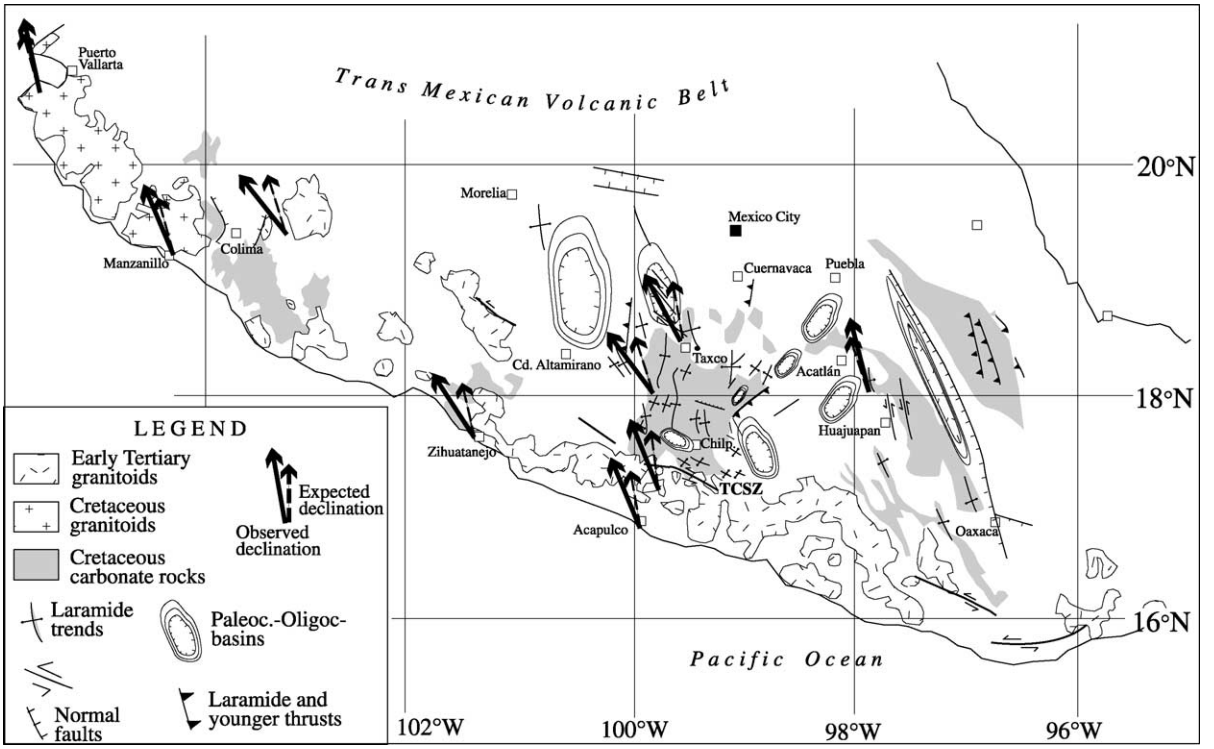


Fig. 8. Schematic tectonic map of southern Mexico showing major Cretaceous elements, Laramide trends, strike-slip faults, Tertiary basins, and paleomagnetically determined rotations. TCSZ = Tierra Caliente shear-zone.

more northerly than those observed closer to the continent margin. Inclinations in Jurassic rocks are not statistically different from inclinations observed in remagnetized Morelos strata and in younger plutons. If our interpretation of these data as Late Cretaceous–Early Tertiary secondary magnetizations is correct, one must conclude that the Laramide foreland of Paleozoic and Precambrian massifs is either not rotated or slightly clockwise rotated with respect to the craton.

7. Conclusions

Paleomagnetic data for the Morelos and Mezcala formations, of the Guerrero–Morelos platform, southern Mexico, indicate that the characteristic magnetization is secondary and was acquired during a Late Cretaceous compressional event. Although reverse polarity (stratigraphically confined) magnetizations were observed in the Morelos Formation, these are also secondary in origin and possibly related to late diagenesis. The overall mean direction indicates that the sampling region has experienced a counterclockwise rotation of $\sim 19 \pm 4.0^\circ$ with respect to the stable craton. Counterclockwise rotations are widespread along the southern Mexico continental margin. These are interpreted to result from interplate stress transmission associated with Tertiary margin truncation and eastward motion of the Chortis and Xolapa blocks.

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References

Aguilera-Franco, N., Hernández-Romano, U., Allison, P.A., 2001. Biostratigraphy and environmental changes across the Cenomanian–Turonian boundary, southern Mexico. J. South Am. Earth Sci. 14, 237–255.

Alaniz-Álvarez, S.A., Nieto-Samaniego, A.F., Morán-Zenteno, D.J., Alba-Aldave, L., 2002. Rhyolitic volcanism in extension zone associated with strike-slip tectonics in the Taxco region, Southern México. J. Volcanol. Geotherm. Res. 118, 1–14.

Alvarez, W., Arthur, M.A., Fischer, A.G., Lowrie, W., Napoleone, G., Premoli-Silva, I., Roggenthen, W.M., 1977. Upper Cretaceous–Paleocene magnetostratigraphy at Gubbio, Italy: V. Type section for the Late Cretaceous–Paleocene geomagnetic reversal time scale. Geol. Soc. Am. Bull. 95, 324–336.

Böhnel, H., 1999. Paleomagnetic study of Jurassic and Cretaceous rocks from the Mixteca terrane (Mexico). J. South Am. Earth Sci. 12, 545–556.

Böhnel, H., Negendank, J.F.W., Urrutia-Fucugauchi, J., 1988. Paleomagnetism and ore petrology of three Cretaceous–Tertiary batoliths of southern Mexico. N. Jb. Geol. Paläont. Mh. 2, 97–127.

Böhnel, H., Alva-Valdivia, L., González-Huesca, S., Urrutia-Fucugauchi, J., Morán-Zenteno, D., Schaaf, P., 1989. Paleomagnetic data and the accretion of the Guerrero terrane, southern Mexico continental margin. In: Hillhouse, J.W. (Ed.), Deep Structure and Past Kinematic of Accreted Terranes. AGU Geophysical Monograph/I.U.G.G., vol. 5, pp. 73–92.

Böhnel, H., Gose, W.A., Testarmata, M.M., Bocanegra-Noriega, G., 1990. Paleomagnetic results from Jurassic and cretaceous marine sediments from the southern part of the Sierra Madre Oriental (Mexico). Phys. Earth Planet. Int. 64, 237–246.

Cabral-Cano, E., Lang, H.R., Harrison, C.G.A., 2000. Stratigraphic assessment of the Arcelia–Teloloapan area, southern Mexico: implications for southern Mexico's post-Neocomian tectonic evolution. J. South Am. Earth Sci. 13, 443–457.

Campa, M.F., Coney, P.J., 1983. Tectono-stratigraphic terranes and mineral resource distributions in Mexico. Can. J. Earth Sci. 20, 1040–1052.

Castro Mora, J., Schlaepfer, C.J., Martínez-Rodríguez, E., 1975. Estratigrafía y microfacies del Mesozoico de la Sierra Madre del Sur, Chiapas. Bol. Assoc. Mex. Géologos Petrol. 85, 607–618.

Cerca Martinez, M., Ferrari, L., 2001. Vertically decoupled Paleocene–Eocene wrenching in southern Mexico and its possible relation with the activation of the Caribbean–North America boundary. Eos Trans. AGU 82 (47) (Fall Meet. Suppl., Abstract T12D-0938).

De Cserna, Z., Ortega-Gutiérez, F., Palacios-Nieto, M., 1980. Reconocimiento geológico de la parte central de la cuenca del Alto Río Balsas, Estados de Guerrero y Puebla, Libro-Guía, Excursión Geol. III Reunión Nacional Geotectonia-Geotermia, México. 40 pp.

Elías-Herrera, M., Sánchez-Zavala, J.L., Macías-Romo, C., 2000. Geologic and geochronologic data from the Guerrero terrrane in the tejupilco area, southern Mexico: new constraints on its tectonic interpretation. J. South Am. Earth Sci. 13, 355–375.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond., A 217, 295–305.

Fries, C., 1960. Geología del Estado de Morelos y de partes adyacentes de México y Guerrero, región central-meridional de México. Bol. Inst. Geol., UNAM 60 (236 pp.).

Helsley, C.E., Steiner, M.B., 1969. Evidence of long intervals of normal polarity during the cretaceous period. Earth Planet. Sci. Lett. 5, 325–332.

Hernández-Romano, U., Aguilera-Franco, N., Martínez-Medrano, M., Barcel-Duarte, J., 1997. Guerrero–Morelos platform drowning at the Cenomanian–Turonian boundary, Huitziltepec area, Guerrero State, southern Mexico. Cretac. Res. 18, 661–686.

Herrmann, U.R., Nelson, B.K., Ratschbacher, L., 1994. The origin of a terrane: U/Pb zircon geochronology and tectonic evolution of the xolapa complex (southern Mexico). Tectonics 13, 455–474.

Jones, D.M., Jackson, P.R., 1999. Geologic setting of skarn associated gold deposits of the Nukay district, Guerrero, Mexico. Associación de Ingenieros de Minas, Metalurgistas y Geólogos de México, XXIII Annual Meeting Proceedings, Acapulco, Mexico, pp. 42–46.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data: examples from Siberia and Morroco. Geophys. J. R. Astron. Soc. 62, 699–718.

Kodama, K.P., 1986. Effect of deformation on the fold test. EOS Trans. Am. Geophys. Union 67, 268.

Lang, H.R., Frerichs, W.E., 1998. New planktic foraminiferal data documenting coniacian age for laramide orogeny onset and paleooceanography in southern Mexico. J. Geol. 106, 635–640.

MacDonald, W.D., 1980. Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies. J. Geophys. Res. 85, 3659–3669.

McElhinny, M.W., McFadden, P.L., 2000. Paleomagnetism—Continents and Oceans. Internat. Geophys. Series, vol. 73. Academic Press, London. 386 pp.

McFadden, P.L., 1990. A new fold test for paleomagnetic studies. Geophys. J. Int. 103, 163–169.

McFadden, P.L., Jones, D.L., 1981. The fold test in paleomagnetism. Geophys. J. R. Astron. Soc. 67, 53–58.

Meschede, M., Frisch, W., 1998. A plate-tectonic model for the mesozoic and early cenozoic history of the Caribbean plate. Tectonophysics 296, 269–291.

Molina Garza, R.S., Urrutia-Fucugauchi, J., Van der Voo, R., 1994. Concordant paleomagnetic directions from the Tehuantepec Isthmus: constraints on the Neogene evolution of the North America–Caribbean plate boundary. Tectonophysics 235, 339–346.

Morán-Zenteno, D.J., Urrutia-Fucugauchi, J., Böhnel, H., González-Torres, E., 1988. Paleomagnetismo de rocas Jurasicas del norte de Oaxaca y sus implicaciones tectónicas. Geofís. Int. 27, 485–518.

Morán-Zenteno, D.J., Corona-Chávez, P., Tolson, G., 1996. Uplift

and subduction erosion in southwestern Mexico since the oligocene: pluton geobarometry constraints. Earth Planet. Sci. Lett. 141, 51–65.

Nowicki, M.J., Hall, S.A., Evans, I., 1993. Palaeomagnetic evidence for local and regional post-Eocene rotations in northern Mexico. Geophys. J. Int. 114, 63–75.

Opdyke, N.D., Channell, J.E.T., 1996. Magnetic Stratigraphy, International Geophysics Series, vol. 64. Academic Press, San Diego, CA. 346 pp.

Ortega-Guerrero, B., Urrutia-Fucugauchi, J., 1993. Jurassic paleomagnetic results constraining southerly motions of the mixteca terrane, southern México. Geofís. Int. 32, 475–485.

Ortega-Gutiérrez, F., 1980. Rocas volcánicas del Maestrichtiano en el área de San Juan Tetelcingo, Estado de Guerrero. Libro-Guía, Excursión Geol., vol. III. Reunión Nacional Geotectonia-Geotermia, México, pp. 34–38.

Pantoja-Alor, J., 1983. Geocronometría del magmatismo Cretácico-Terciario de la Sierra Madre del Sur. Abstracts Sociedad Geológica Mexicana, México DF, XLIV, 29.

Tardy, M., Lapierre, H., Freydier, C., et al., 1994. The Guerrero suspect terrane (western Mexico) and coeval arc terranes (the Greater Antilles and the Western Cordillera of Colombia); a late Mesozoic intra-oceanic arc accreted to cratonal America during the cretaceous. Tectonophysics 230, 49–73.

Treviño-Rodríguez, A.F., 1986. Investigaciones magnetostratigráficas y paleomagnéticas de la Formación Morelos, Estado de Guerrero, México. Master Thesis, Universidad Nacional Autónoma de México (unpublished), México. 95 pp.

Urrutia-Fucugauchi, J., 1980. Paleomagnetic studies of Mexican rocks. PhD thesis, Univ. New Castle upon Tyne, UK. 689 pp.

Urrutia-Fucugauchi, J., 1983. Preliminary paleomagnetic study of lower tertiary volcanic rocks from Morelos and Guerrero States. Geofís. Int. 22, 87–110.

Urrutia-Fucugauchi, J., 1988. Paleomagnetic study of the Cretaceous Morelos Formation, Guerrero state, southern Mexico. Tectonophysics 147, 121–125.

Urrutia-Fucugauchi, J., Molina-Garza, R., 1995. Magnetic fabrics and rock magnetism of the Xaltianguis intrusive, southern Mexico: implications for emplacement mode. Geofís. Int. 34, 201–211.

Urrutia-Fucugauchi, J., Morán-Zenteno, D.J., Cabral Cano, E., 1987. Paleomagnetism and tectonics of Mexico. Geofís. Int. 26, 429–458.

Van der Plujim, B.A., 1987. Grain-scale deformation and the fold test-evaluation of syn-folding remagnetization. Geophys. Res. Lett. 14, 155–157.

Van Hinte, J.E., 1976. A Jurassic and Cretaceous time scale. Am. Assoc. Pet. Geol. Bull. 60, 489–516.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W, Creer, K.M., Runcorn, S.K. (Eds.), Methods in Rock Magnetism and Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.